

TIRANA RINAS, Albania

WMO: 136150

Lat: 41.415N

Lon: 19.721E

Elev: 126

StdP: 14.63

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.0	28.7	10.8	9.6	38.8	16.7	12.8	38.9	21.8	52.2	18.6	51.1	2.6	170	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.3	95.1	72.5	92.9	72.9	90.8	73.3	81.2	85.9	79.0	84.5	77.0	84.0	7.5	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	159.2	83.8	77.4	143.0	81.7	75.3	133.1	79.7	45.0	87.0	42.7	83.1	40.7	82.3	91.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.0	14.4	12.2	DB	20.8	99.8	3.6	3.1	18.2	102.0	16.1	103.8	14.0	105.5	11.4	107.7
			WB	19.7	82.6	4.0	5.2	16.9	86.3	14.5	89.4	12.3	92.3	9.4	96.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.2	44.7	47.9	52.6	58.9	65.4	73.1	77.4	78.1	70.8	62.7	54.9	46.7	(6)
	DBStd	12.56	6.07	5.89	4.43	4.28	4.82	5.25	3.61	3.46	5.02	5.23	5.92	6.18	(7)
	HDD50	467	178	99	25	1	0	0	0	0	0	1	23	140	(8)
	HDD65	2724	628	480	383	189	56	4	0	0	11	102	305	566	(9)
	CDD50	4550	16	40	107	268	476	693	850	871	625	396	170	38	(10)
	CDD65	1330	0	0	0	5	67	247	385	406	186	32	2	0	(11)
	CDH74	13398	0	2	3	83	583	2413	4126	4302	1625	253	8	0	(12)
	CDH80	5905	0	0	0	10	123	983	1986	2194	584	24	1	0	(13)
(14) Wind	WSAvg	4.6	4.7	5.5	5.5	4.6	4.6	4.5	4.6	4.5	4.2	3.8	4.2	4.5	(14)
(15) Precipitation	PrecAvg	35.9	3.0	3.2	3.6	3.6	1.9	1.8	1.2	1.3	3.2	3.7	5.0	4.6	(15)
	PrecMax	51.1	8.0	8.1	7.2	6.6	4.5	5.0	3.5	6.1	11.0	9.0	11.4	10.6	(16)
	PrecMin	21.1	0.4	0.0	0.1	0.6	0.7	0.0	0.0	0.0	0.0	0.0	1.9	0.7	(17)
	PrecStd	7.9	2.0	2.1	2.2	1.7	1.1	1.5	0.9	1.5	2.6	2.3	2.4	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.7	69.5	73.1	81.4	87.5	95.3	98.2	98.8	93.1	82.8	74.9	66.6	(19)
		MCWB	53.8	54.1	54.8	64.1	72.1	73.7	72.2	71.7	70.2	65.4	64.1	60.3	(20)
	2%	DB	60.7	64.6	69.5	75.8	83.7	91.7	94.8	95.4	89.8	79.5	71.2	62.8	(21)
		MCWB	53.0	54.2	55.2	62.2	71.5	75.1	73.4	71.8	70.0	65.8	61.2	56.5	(22)
	5%	DB	57.8	61.2	66.3	73.0	80.3	89.4	92.4	93.3	86.4	76.9	68.4	60.4	(23)
		MCWB	51.9	52.7	55.3	60.5	68.8	74.4	73.5	72.1	69.7	65.7	59.2	54.0	(24)
	10%	DB	55.3	58.7	64.0	70.0	76.9	86.2	90.0	91.1	83.0	73.8	65.9	57.6	(25)
		MCWB	49.9	51.0	54.0	59.8	66.2	73.1	73.6	73.1	68.8	64.7	58.6	52.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.6	60.9	63.9	69.5	78.0	85.5	85.4	83.4	78.9	75.0	67.9	63.0	(27)
		MCDB	60.6	62.4	65.1	71.9	82.3	89.1	87.8	87.6	82.2	76.8	70.0	64.8	(28)
	2%	WB	56.0	57.2	60.0	66.7	74.5	80.7	81.3	80.0	75.6	71.6	64.2	59.1	(29)
		MCDB	57.6	60.7	62.6	70.6	79.2	85.9	86.0	85.6	81.3	73.8	67.2	60.7	(30)
	5%	WB	53.3	54.6	57.5	64.4	71.7	77.5	78.9	78.0	73.2	68.2	61.8	56.1	(31)
		MCDB	55.8	59.6	62.6	69.5	76.9	83.4	84.8	85.0	79.9	72.4	65.4	58.7	(32)
	10%	WB	51.1	52.3	55.6	62.2	68.7	74.8	76.8	76.0	71.0	65.9	59.8	53.6	(33)
		MCDB	54.4	57.4	61.3	67.7	74.4	81.9	84.0	84.2	78.7	71.6	63.9	55.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	18.8	20.3	20.7	20.7	21.4	23.9	25.3	23.3	22.1	19.8	20.0	(35)
		MCDBR	21.4	22.3	25.3	26.0	25.1	25.7	27.4	28.9	27.0	25.5	22.0	21.4	(36)
	5% WB	MCWBR	14.8	13.7	14.4	14.7	14.1	12.9	11.7	11.1	11.8	14.0	13.4	14.7	(37)
		MCWBR	17.7	18.5	20.5	22.4	22.9	22.7	23.3	24.0	23.2	21.1	18.8	16.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.346	0.373	0.421	0.475	0.463	0.439	0.441	0.460	0.439	0.406	0.374	0.348	(40)
		taud	2.410	2.313	2.178	2.047	2.149	2.272	2.259	2.199	2.246	2.314	2.357	2.404	(41)
	Ebn at Noon		252	261	260	253	260	266	265	255	252	246	238	240	(42)
		Edh at Noon	26	33	42	51	48	42	42	44	39	33	27	24	(43)
(44) All-Sky Solar Radiation	RadAvg		564	820	1209	1575	1965	2260	2336	2046	1487	1027	650	500	(44)
	RadStd		59	119	158	160	141	131	95	106	108	96	60	84	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-144	N/A	N/A

Nomenclature: See separate page